

Class 27: Change of Basis

Review: If $\mathbf{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_k\}$ is a basis for vector space V and $\mathbf{x}$ is any vector in V we can write $\mathbf{x}$ as a unique linear combination of the basis vectors. The weights in the linear combination are the $\mathbf{B}$ -coordinates of $\mathbf{x}$.

Example in $\mathbf{R}^3$. Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$
Find $[\mathbf{x}]_{\mathcal{B}}$.

In general $\mathbf{P}_{\mathcal{B}} [\mathbf{x}]_{\mathcal{B}} = \mathbf{x}$ in standard coordinates.

Now if $\mathbf{C}$ is another basis for V the vector $\mathbf{x}$ will have a different set of coordinates with respect to the $\mathbf{C}$ basis.

The goal for today is to discover how to convert the $\mathbf{B}$ coordinate vector, $[\mathbf{x}]_{\mathcal{B}}$, directly into the $\mathbf{C}$ coordinate vector $[\mathbf{x}]_{\mathcal{C}}$.

We know for bases $\mathbf{B}$ and $\mathbf{C}$ and any vector $\mathbf{x}$ we have

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To convert $[\mathbf{x}]_{\mathbf{B}}$ to $[\mathbf{x}]_{\mathbf{C}}$ we simply multiply by a matrix. This matrix is

Our book denotes this matrix by

We find this change of coordinates matrix by row reducing as follows.

Key idea: This change of coordinates is a linear transformation since it is accomplished by simply multiplying by a matrix.

Example: Let $B = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \end{bmatrix} \right\}$ and let $C = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

Find the coordinates of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}_B$ relative to basis **C**.

What is the vector in standard coordinates?

What is the matrix that converts **C** coordinates to **B** coordinates?

